Pipeline Resilience and Fault-Tolerance Matrix

This document provides a highly structured, user-friendly mapping of all pipeline steps grouped by their execution classes and concurrency profiles. Symmetrically, this matrix specifies how each group behaves under our 3-Category API Resiliency Taxonomy when encountering transient capacity spikes (Category A), unrecoverable timeouts (Category B), or configuration errors (Category C).

Agentic Loops (Thread-Separated Resilient Loops)

These are multi-turn, conversational, agentic loops operating concurrently inside a parallel thread pool (where each document page or macro-view runs as an independent worker thread). Symmetrically, Category B and C exceptions are caught locally inside the thread to execute census rollbacks on disk, allowing other concurrent workers to run to completion and serialize successfully:

Pipeline Step Name	Category A (503/502) Resilience	Category B (504/429/500/JSONDecodeError) Resilience	Category C (400/403/404) Resilience	Exit & Fail-Fast Mechanics
Macro-Discovery Loop	3 attempts (Jittered Backoff)	Worker-isolated trapping, page census checklist rollback to `PENDING`	Worker-isolated trapping, page census checklist rollback to `PENDING`	**Resilient Workers:** Parallel threads complete successfully; round completes with orange warnings
Micro-Subdivision Loop	3 attempts (Jittered Backoff)	Worker-isolated trapping, page census checklist rollback to `PENDING`	Worker-isolated trapping, page census checklist rollback to `PENDING`	**Resilient Workers:** Parallel threads complete successfully; round

Pipeline Step Name	Category A (503/502) Resilience	Category B (504/429/500/JSONDecodeError) Resilience	Category C (400/403/404) Resilience	Exit & Fail-Fast Mechanics
				completes with orange warnings
Decoupled Verification Loop	3 attempts (Jittered Backoff)	Worker-isolated trapping, page census checklist rollback to `PENDING` (triggered on 504 Deadline Exceeded or JSONDecodeError anomalies)	Worker-isolated trapping, page census checklist rollback to `PENDING`	**Resilient Workers:** Parallel threads complete successfully; round completes with orange warnings

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Standard Parallel Steps (Fail-Fast Concurrently)

These are concurrent bulk-operations executing inside a thread pool, but do not represent independent page-agent conversational loops. Symmetrically, they implement a hybrid concurrency model distinguishing systemic configuration errors from transient server-side overloads to maximize overall pipeline throughput:

Pipeline Step Name	Category A (503/502) Resilience	Category B (504/429/500) Resilience	Category C (400/403/404) Resilience	Exit & Fail-Fast Mechanics
Document Layer Proposal	3 attempts (Jittered Backoff)	Local-isolated trapping, increment errors, print warning	Local-isolated trapping, increment errors, print warning	**Resilient Sequential:** Symmetrically isolates failures per document to let others complete; completes with orange completed-warning

Pipeline Step Name	Category A (503/502) Resilience	Category B (504/429/500) Resilience	Category C (400/403/404) Resilience	Exit & Fail-Fast Mechanics
Asset Layer Assignment	3 attempts (Jittered Backoff)	Worker-isolated trapping, print orange System Warning alert, halt that worker only	Immediate thread pool Early-Abort	**Hybrid Resilience:** Symmetrically early-aborts on Category C, but isolates Category B locally to let siblings complete (printing orange warnings)
Asset Captioning Step	3 attempts (Jittered Backoff)	Worker-isolated trapping, print orange System Warning alert, halt that worker only	Immediate thread pool Early-Abort	**Hybrid Resilience:** Symmetrically early-aborts on Category C, but isolates Category B locally to let siblings complete (printing orange warnings)

Hybrid Decoupled Indexing (Parallel Batch Vectorization)

These are decoupled indexing channels that separate asynchronous network-bound translation threads (MIMD) from C++ model matrix vectorizations (SIMD) under global concurrency locks:

Pipeline Step Name	Category A (503/502) Resilience	Category B (504/429/500) Resilience	Category C (400/403/404) Resilience	Exit & Fail-Fast Mechanics
Index Building Step	3 attempts (Jittered Backoff)	Batch-isolated trapping, document rollback, print orange warning	Immediate process halt	**Decoupled Resilience:** Symmetrically early-aborts on Category C and exits, but isolates Category B per-batch to roll back partial index

Pipeline Step Name	Category A (503/502) Resilience	Category B (504/429/500) Resilience	Category C (400/403/404) Resilience	Exit & Fail-Fast Mechanics
				writes and preserve FTS/Chroma files for completed documents.

Single-Thread Commands (Fail-Fast Sequentially)

These sequential steps run strictly on the main thread (exactly 1 thread) with zero thread-separation. Symmetrically, any Category B or C error immediately halts execution, aborts further processing, and maps to our unclassified red failure milestones and process-level exits:

Pipeline Step Name	Category A (503/502) Resilience	Category B (504/429/500) Resilience	Category C (400/403/404) Resilience	Exit & Fail-Fast Mechanics
Single-Query Step	3 attempts (Jittered Backoff)	Immediate process halt	Immediate process halt	**Fail-Fast Transactional:** Main thread halts instantly, propagates synthesis exceptions, and exits with red failure milestone
Batch-Query Step	3 attempts (Jittered Backoff)	Query-isolated trapping, write error to report, continue	Query-isolated trapping, write error to report, continue	**Resilient Batch:** Symmetrically isolates failures per query to let others complete; writes final markdown report

Pipeline Step Name	Category A (503/502) Resilience	Category B (504/429/500) Resilience	Category C (400/403/404) Resilience	Exit & Fail-Fast Mechanics
Conversational REPL Session	3 attempts (Jittered Backoff)	Local-turn interception, continue loop	Immediate process halt	**Fail-Fast REPL:** Main thread halts instantly on startup failures, exits with preflight code, and intercepts turn-level drops inline

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Local-First Commands (Fully Offline Operations)

These sequential steps perform critical local-first technical validations and compilations on disk. Symmetrically, they never make any Generative API calls or remote network handshakes, meaning Category A, B, and C errors are mathematically impossible:

Pipeline Step Name	Category A (503/502) Resilience	Category B (504/429/500) Resilience	Category C (400/403/404) Resilience	Exit & Fail-Fast Mechanics
Document Ingestion Step	N/A (Fully Offline)	N/A (Fully Offline)	N/A (Fully Offline)	**Fail-Fast Systemic:** Local filesystem or permission errors halt main thread and exit immediately
Deep-Integrity Verification	N/A (Fully Offline)	N/A (Fully Offline)	N/A (Fully Offline)	**Gated Warnings:** Failed checks report orange completed warning; script crashes exit immediately
Package Exporting Step	N/A (Fully Offline)	N/A (Fully Offline)	N/A (Fully Offline)	**Gated Warnings:** Missing success marker reports orange warning; script crashes exit immediately